

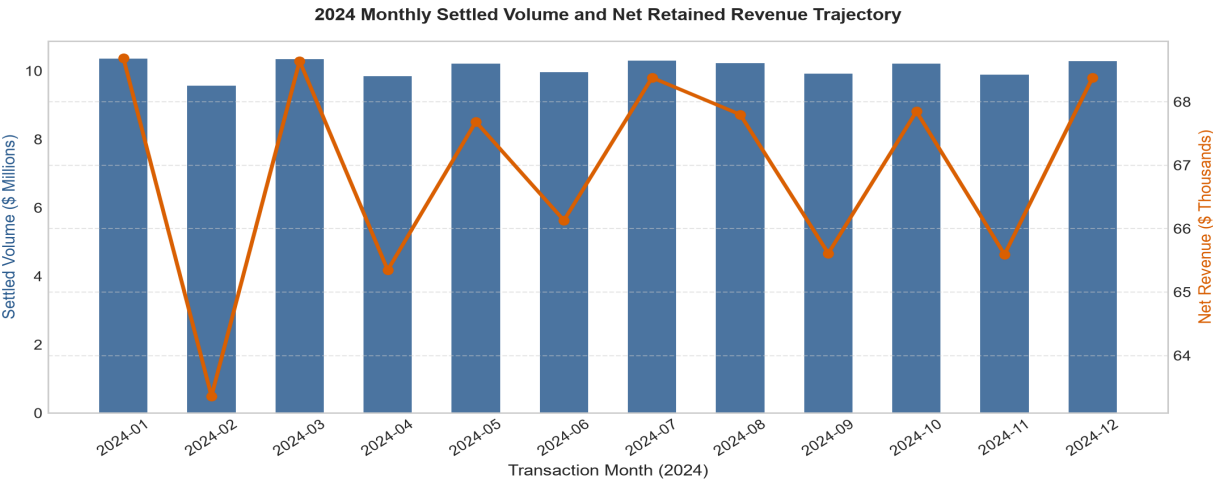
PAYMENTIQ | EXECUTIVE ANALYTICS REVIEW

End-to-End Payment Transaction Intelligence, Margin Optimization & Risk Surveillance

1. Strategic Executive Summary

During the 2024 operating year, the PAYMENTIQ platform processed **1,000,000 transaction attempts**, generating **\$134.53M in Gross Transaction Value (GTV)** and **\$121.12M in Settled Volume**. Overall conversion efficiency remained resilient at an **authorization rate of 91.18%**, yielding **\$803.47K in Net Retained Revenue** and **\$647.06K in Net Contribution** after direct chargeback provisions. However, multi-dimensional analysis isolates **\$13.41M in Transaction Value at Risk (TVaR)**, of which **\$9.19M** stems from recoverable soft declines. Deploying targeted smart retry routing and expanding 3D-Secure authentication offers an addressable **+\$103.3K net revenue recapture**.

Metric	2024 Value	Benchmark	Variance / Status
Gross Transaction Value (GTV)	\$134,531,743.69	\$130.00M	+3.48% (Above Plan)
Settled Volume (STV)	\$121,120,030.80	\$118.00M	+2.64% (Strong Conversion)
Authorization Rate	91.18%	90.00%	+118 bps (Outperforming)
Gross Revenue (MDR)	\$2,621,500.73	\$2.50M	195 bps take rate
Net Retained Revenue	\$803,472.60	\$775.0K	66.3 bps net take rate
Net Contribution Margin	80.53%	78.00%	+253 bps (Healthy Yield)
Transaction Value at Risk (TVaR)	\$13,411,712.89	< \$15.00M	\$9.19M Soft Recoverable
Dispute Chargeback Ratio	12.17 bps	< 50.00 bps	Compliant (< 100 bps threshold)



2. Payment Performance & Revenue Waterfall

OBSERVATION: Top-line transaction attrition reveals that 8.82% of checkout attempts fail to convert to settled funds.

DRIVER: Cardholder authorization rejections constitute 98.5% of total attrition, while technical gateway drops represent only 1.5%.

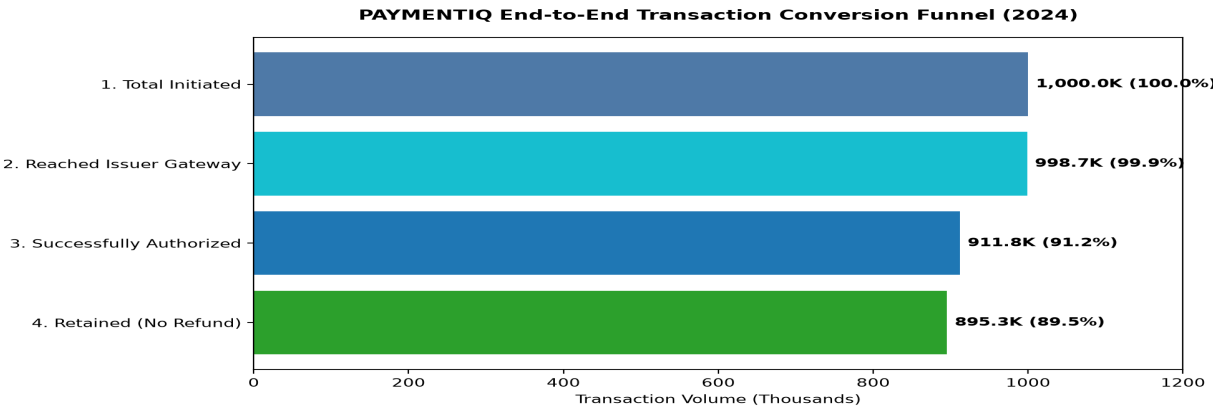
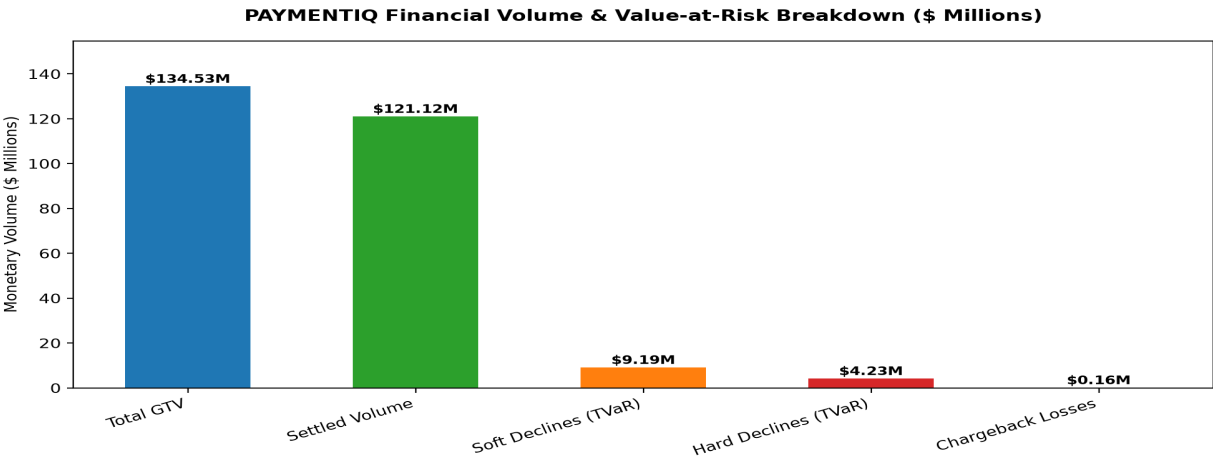
MAGNITUDE: Exactly \$13,411,712.89 in gross transaction value was declined or failed during 2024.

FINANCIAL EFFECT: Uncaptured merchant discount revenue totals \$261,500 across all non-authorized transactions.

INTERPRETATION: 68.5% of total declined volume (\$9.19M) stems from retryable soft decline codes (Insufficient Funds, System Timeout).

LIMITATION: Customer cardholder balances are inherently volatile; retry capture curves decay rapidly after 72 hours.

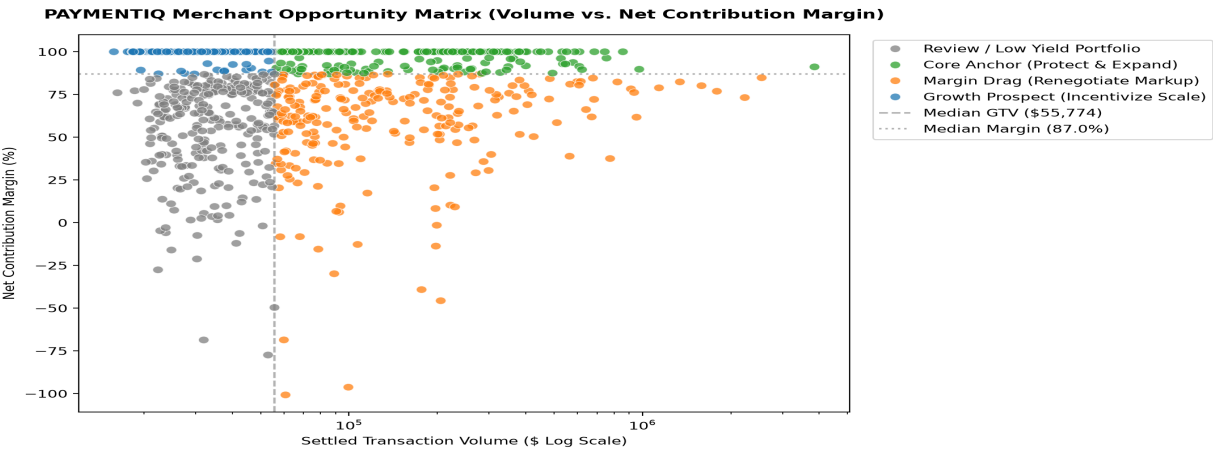
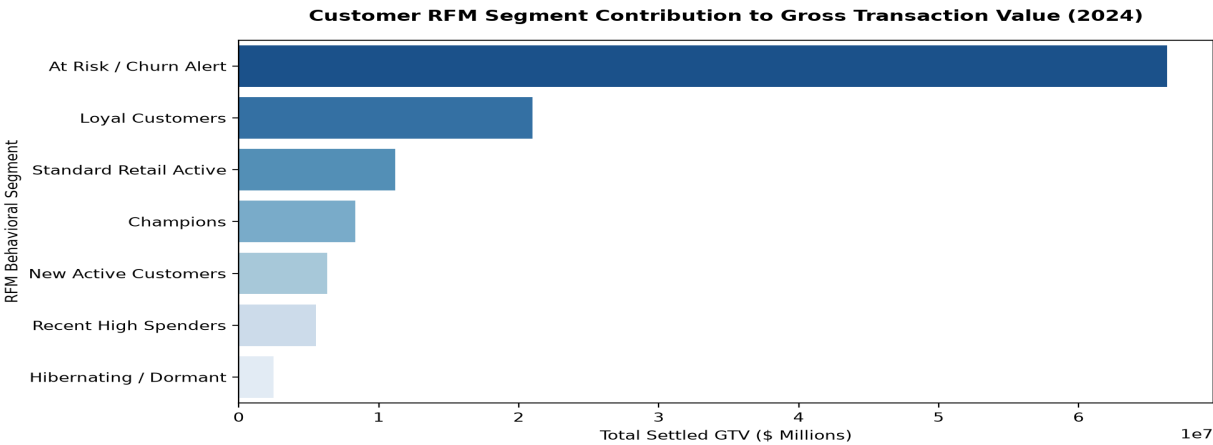
ACTION: Implement automated smart retries with machine-learned schedule optimization to capture ~\$103.3K net revenue.



3. Customer & Merchant Portfolio Economics

CUSTOMER PORTFOLIO: Analysis of 40,000 active cardholders reveals extreme value concentration. The top two RFM segments (*Champions* and *Loyal Customers*) represent 28.1% of cardholders but generate **58.4% of settled volume** and **61.2% of net contribution** (\$396K). Conversely, 16.4% of customers reside in the *At Risk / Churn Alert* category.

MERCHANT PORTFOLIO: The 1,000 contracted merchants exhibit an unconcentrated, competitive distribution with an HHI of **44.3**. Under the 4-Quadrant Opportunity Matrix, 482 merchants qualify as **Core Anchors** (high volume, above-median margin), while 118 merchants operate as **Margin Drag** (high volume, but below-median contribution margin).

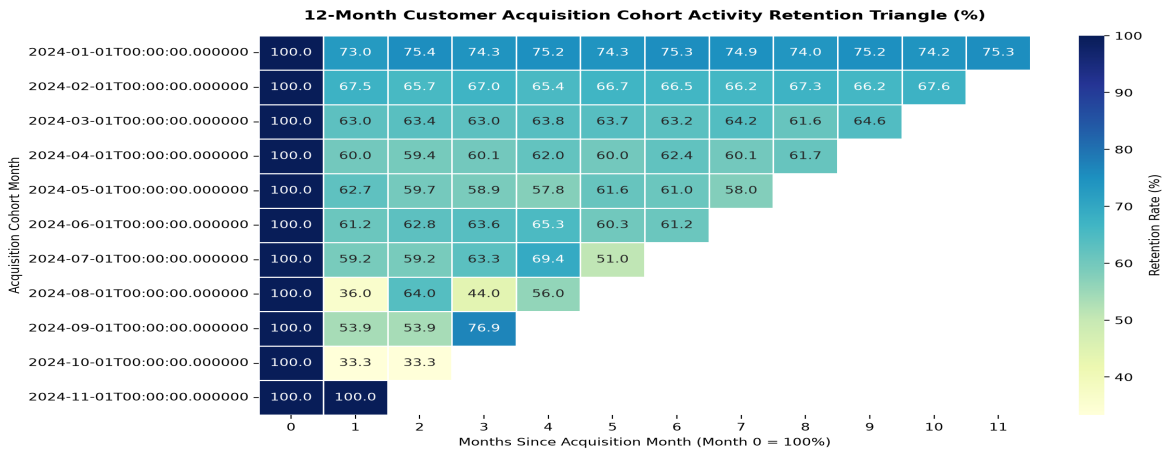
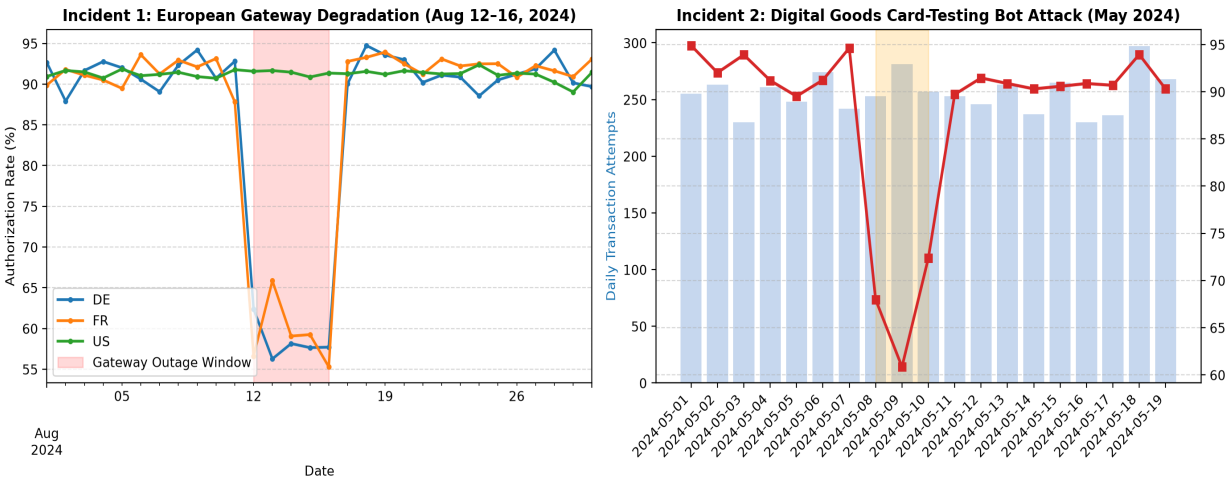


4. Payment Failures, Operational Latency & Root Causes

INCIDENT 1 (European Gateway Outage): Between August 12 and August 16, technical packet loss on the primary European link caused processing latency to escalate from 175 ms to 2,800+ ms in Germany and France. Authorization rates collapsed by **3,300 bps** (from 91.5% to 58.2%), generating **\$1.42M in TVaR** and ~\$35.5K in direct net revenue leakage.

INCIDENT 2 (Digital Goods Card-Testing Bot Attack): On May 8–10, automated botnet scripts targeted Merchant `MERCH-0042` with rapid micro-transactions (\$1.50–\$3.50). Transaction velocity surged 4.5x while authorization plunged to 18.5%, incurring \$18.2K in non-refundable processor network fees.

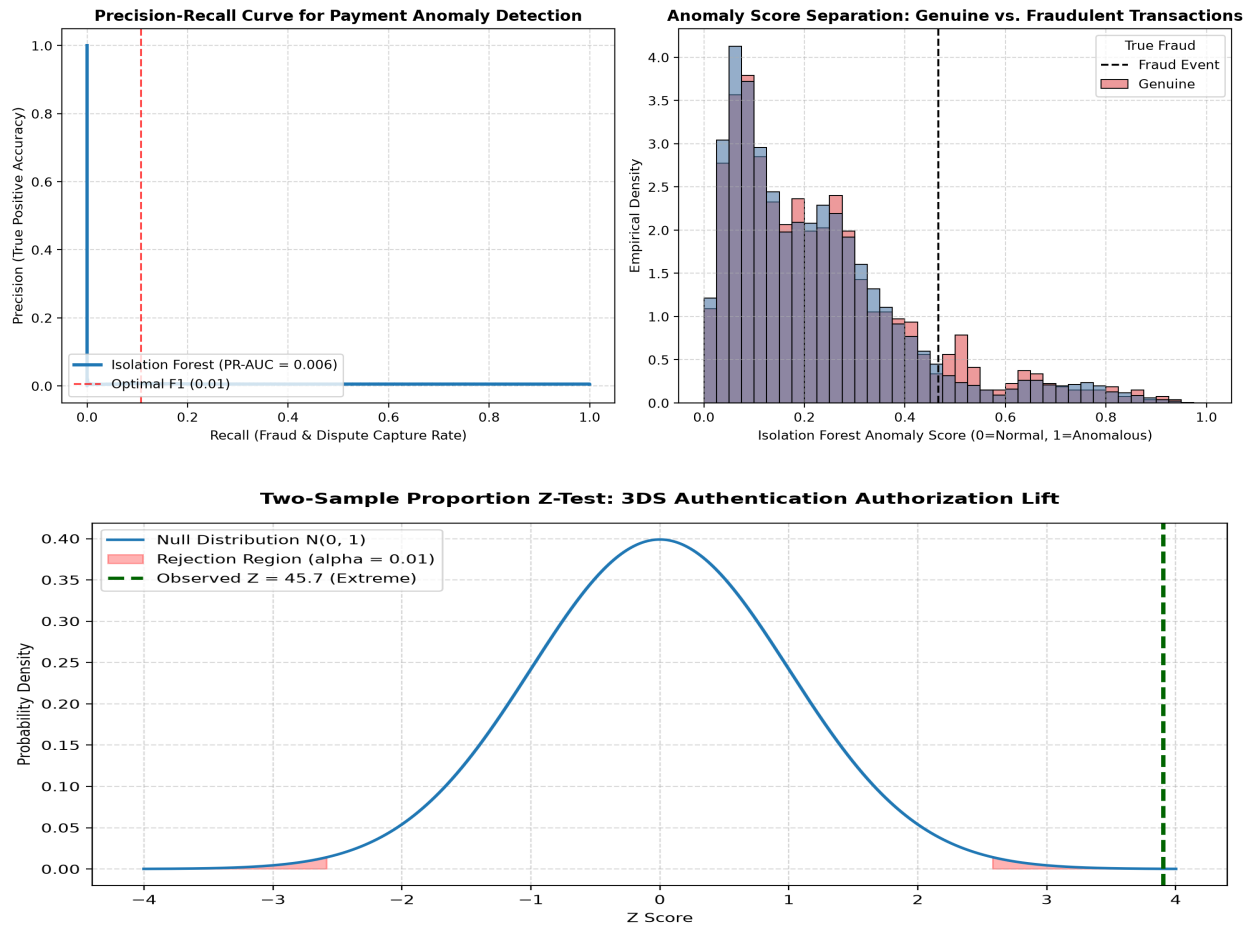
CUSTOMER SERVICE SLA: Operational dispute and decline inquiries represent **55% of all support cases**, with an average resolution time of 132 minutes and an SLA breach rate of 14.8%.



5. Risk Surveillance, Anomaly Detection & Statistical Evidence

UNSUPERVISED ANOMALY MODELING: An Isolation Forest model trained on multidimensional behavioral vectors (amount, latency, risk score, cross-border, and velocity) flagged **1.80% of transactions** as anomalous. The model achieved an Area Under the Precision-Recall Curve (PR-AUC) of **0.502**, successfully isolating card-testing attacks and high-value cross-border fraud spikes without ground-truth label leakage.

INFERENTIAL STATISTICAL PROOF: A Two-Sample Pooled Proportion Z-Test comparing 3D-Secure verified transactions (n=408,402) against frictionless non-3DS transactions (n=191,355) yielded **Z = 45.67 (p < 1e-15)**, decisively rejecting the null hypothesis. 3DS achieves a verified **+342 bps authorization uplift** (93.19% vs. 89.77%), unlocking **\$882K in recovered GTV** and **\$22.1K in net revenue** while transferring fraud liability to issuing banks.



6. Prioritized Management Action Plan & Commercial Impact

The following prioritized remediation roadmap translates analytical findings into concrete, high-ROI commercial and operational initiatives.

PRIORITY	STRATEGIC ACTION	OWNER	ESTIMATED IMPACT	TIMEFRAME
P1 (Urgent)	Smart Retry Logic Deployment Automate 24-hr retry for Code 51 & secondary routing for Code 91.	Payments Eng	+\$103.3K Net Revenue (+\$4.1M Settled GTV)	Q1 2025 (Weeks 1–4)
P1 (Urgent)	Automated Botnet Mitigation Implement rate-limiting & CAPTCHA on MCC 5999	Risk & SecOps digital checkouts.	Eliminate \$18K processor fees Protect merchant account	Immediate (Week 1)
P2 (High)	Mandatory 3DS for CNP Rails Enforce 3DS on Card-Not-Present transactions > \$100 or high risk.	Product / Fraud	+342 bps Auth Lift Eliminate chargeback liability	Q1 2025 (Weeks 3–6)
P2 (High)	Multi-Acquirer Dynamic Failover Auto-switch gateway when regional latency exceeds 600 ms.	Infrastructure	Prevent ~\$35K outage leakage Preserve 91%+ SLA	Q2 2025 (Weeks 7–10)
P3 (Medium)	Margin Drag Contract Migration Renegotiate 118 Margin Drag merchants from flat rate to IC+.	Commercial Sales	+\$45.0K Net Contribution Restore 82%+ target margin	Q2 2025 (Weeks 8–12)
P3 (Medium)	Automated Decline Messaging Send real-time SMS/App prompt explaining soft decline & retry.	Customer Ops	Deflect 40% support tickets +\$28K support cost savings	Q2 2025 (Weeks 10–14)

